

Enriched strawberry yield through UAV and Landsat fusion, incorporating stacked regressions

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Abstract—This document is a research proposal for CS4610. The proposed method will (1) co-register UAV, Landsat, and DEM data at several candidate common resolutions, (2) derive spectral (bands, NDVI) and terrain (slope, aspect, ruggedness) predictors, and (3) train and evaluate multiple regression and stacking-ensemble models to predict UAV-derived NDVI from Landsat and DEM features. By comparing performance across resolutions, the study aims to identify an optimal fusion scale and demonstrate that regression-based fusion can enhance spatial detail without sacrificing spectral consistency, enabling more informative, large-area suitability mapping.

I. INTRODUCTION

Crop suitability modeling is the process of evaluating how appropriate a specific area of land is for growing particular crops. This technique is utilized in geographic information systems (GIS) to enhance agricultural productivity by considering environmental, climatic, and soil characteristics. By utilizing modern data sources and methods such as remote sensing and machine learning, farmers and land planners can more accurately assess which crops are suitable. The use cases span from crop monitoring, nutrition assessment, water demand, and pest/disease identification [2].

High-resolution images are essential in crop monitoring because they provide the most accurate and detailed information. Some challenges arise in remote sensing when trying to obtain these high-resolution images. Data from different sources have both strengths and limitations that need to be addressed.

Unmanned Aerial Vehicles (UAVs) excel at capturing finer detail due to their ability to fly low to the ground. These sensors precisely capture features such as soil type, topology, and vegetation. However, due to the manual nature of drone data collection, the area of coverage is very limited and heavily time-intensive. Satellite, on the other hand, such as Landsat, covers vast amounts of areas and images are taken on a regular schedule. Their strength lies in having many more sensors to capture multispectral bands. The only downside is the lower resolution of these images. This data can be difficult to interpret without advanced processing techniques, causing challenges in distinguishing between different crop types due to similarities in their backscatter signatures [2]. Being able to obtain recent, high-resolution images that cover large areas is a problem for farmers. It is too expensive and time-consuming. The solution is being able to fuse these two data sources.

II. MOTIVATION

Modern agriculture faces significant challenges in optimizing crop selection and land use in an evolving landscape of climate change, resource constraints, and the urgent need for sustainable farming. Traditional methods of suitability assessment rely on low-resolution datasets. One of the primary challenges of integrating data from different sensors with different resolutions is ensuring the compatibility of spatial resolutions and characteristics [2]. Creating this new fused data will allow us to extract better insights about the area of land.

III. RESEARCH PROBLEM

Crop suitability modeling is an important tool for farmers and agriculturalists to most efficiently create better yields. However, farmers need accurate, up-to-date high resolution information over a large area. Fusing satellite and UAV data at a common resolution without spectral distortions can improve precision agriculture.

A. Research Objectives

- Fuse satellite and drone images to achieve high-resolution and high-coverage images.
- Use the NDVI values of a high-resolution image to assess crop suitability.
- Test ensemble learning for improving predictions.

IV. LITERATURE REVIEW

A. Classical Image Fusion Algorithms

For this project, we have to consider the tradeoffs associated with different algorithms. The main variables we have to consider are pixel resolution and spectral quality. However, at the same time, we should consider the computational costs associated with certain techniques.

These algorithms are separated into three types: pixel level, feature level, and decision level. The most common type is pixel-level fusion for remote sensing and computer vision applications [2]. Their popularity is for good reason. Pixel-level seeks to achieve both high spatial resolution and spectral resolution. However, they're may be some distortion in the image depending on the method. These are generally fast and easier to implement. The feature level allows the integration of combined features that could improve performance, but requires initial feature extraction procedures. Lastly, the

decision method merges data after sources have undergone classification [6]. This method mainly suffers from complexity and the problem of local maxima.

Mainly, pixel-level fusion algorithms are used, such as Brovey Transform and Principal Component Analysis (PCA). Brovey transform is a multiplicative, ratio-based method that injects high-resolution intensity into each spectral band. It is able to emphasize UAV data, where it provides the most detail. On the other hand PCA algorithm transforms the original spectral data set into a set of components, then fuses them by averaging them and applying a transformation. This method reduces data dimensionally with a good balance of resolution and spectral detail [1].

B. Feature Extraction

Bhandari, Kumar, and Singh present an improved method for the analysis of satellite images based on NDVI. This method calculates NDVI from the separated bands and creates an NDVI image of the observation of vegetation. Based on the wavelengths of NDVI, we determine thresholds that represent different things, like barren rock or grassland. The final output is a false colour composite image that contains colored pixels that represent these features. This method is very useful in detecting surface features of cropland and the visible area [3].

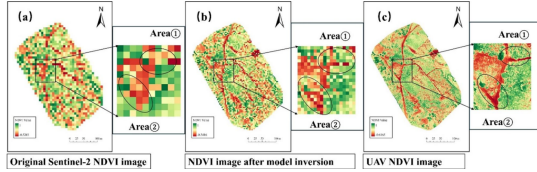


Fig. 1. Original and resampled NDVI images from Sentinel-2 illustrating how higher spatial resolution reveals finer spatial patterns [10].

This is the main vegetation index for monitoring crop growth and health. However, there are many other vegetation indexes that are useful depending on what you are trying to solve. For example, the Normalized Difference Water Index is better used for water stress detection and flood mapping [5].

Digital Elevation Model (DEM) is a 3D mapping of the terrain. This source of data is important in providing specific data, such as elevation, that can give us a better understanding of the terrain. Certain 3D features are important in analyzing crop yield. In a paper researching fusion for better wheat yields, it takes into account the canopy or top layer of the crop. [4] This 3D information is important for understanding how tall the wheat is, hence giving a better idea of the yield. Incorporating DEM features into the model provides a strong, non-noisy predictor. In a different research paper successfully incorporating ML techniques for soybean suitability, they use canopy cover as a feature. This feature is extracted by how much ground is covered by plants. A problem they found is that canopy cover saturates quicker because it doesn't increase much even if the predictors increase [7]. Essentially, hurting model accuracy.

C. Regression Methods

Regression allows us to treat data fusion as a prediction problem. If needed, we can add more predictors to better model the relationship (besides NDVI). This is also important because with regression, we are able to preserve the spectral bands of the data while still sharpening the resolution. While upsampling the resolution of the satellite to a more precise UAV level seems to be the most optimal solution, that may not always be the case.

Then at which level should the fusion be conducted for optimal fusion? Satellite pixels represent the averaged spectral information from multiple ground objects due to their coarse pixel size; hence, it is harder to inject finer detail from the UAV into the satellite. The same applies to the inverse approach. To approach this, we will test multiple candidate resolutions between the native UAV and satellite resolutions, prioritising resolution sharpness. By comparing metrics across different resolutions, we can determine where the model achieves the highest predictive accuracy without generating unrealistic fine-scale detail that is not supported by the satellite image [8].

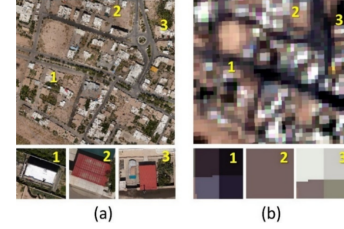


Fig. 2. Different information content and interpretability levels of UAV (a) and Sentinel (b) imageries. This leads to the problem wherein, at a common point on Earth, the UAV-borne sensor observes numerous fine-scale objects while the Sentinel MSI fails to capture any spectral information about them [8].

D. Metaheuristic Approach

One way to find this optimal resolution is to apply a metaheuristic. Or a search strategy for finding good solutions when trying every possible option would take too long. A paper proposes this method when optimizing the fusion of UAV and Sentinel-2 satellite data by using a genetic algorithm as a metaheuristic. To determine a good candidate, the algorithm attempts to optimize an objective function that combines two goals. GRD or ground resolution distance is minimized to ensure the sharpness of images. And Mutual Information (MI) measures the similarity or dependence of information between the rescaled UAV panchromatic band and the Sentinel-2 component. The key to this approach is the efficiency of testing thousands of GRDs through the genetic algorithm [8].

E. Ensemble Learning

Looking to improve the accuracy of results, we can employ ensemble learning. It is an approach of training multiple base learners that are generated from training data, then combining their predictions to get a better overall result than a singular model. The actual generation of learners requires a base

learning algorithm, which can be any other kind of learning algorithm.

Bagging is training many models on bootstrapped samples and then performing an average/vote. The advantage of this approach is that it reduces overfitting and is more immune to noise. Boosting refers to converting weak learners into strong learners. It trains models sequentially, focusing on previous errors and correcting them in the new model. Examples of this algorithm include AdaBoost and XGBoost. Mainly reduces bias and improves accuracy. Stacking is training diverse models and then a separate metamodel that learns how to best combine their predictions. The models can be of many different types and are especially useful when errors are complementary [11].

Due to the relationship of the data, stacking seems the best fit for this research problem. There is a variety of mixed data (spectral bands, NDVI, DEM predictors) with relationships that are likely not to be linear. We want to apply a wide variety of different models too, because the more similar the models, the less advantage there is to stacking. The method of stacking never does worse than selecting the single best predictor [9].

V. TECHNICAL BACKGROUND

A. Resampling Methods

This nearest neighbor method finds training sample pixels closest to a new pixel, predicting the label from these. This method is simple and efficient, allowing the original dataset's spectral features to be preserved. However it can lead to spatial discontinuities especially at edges. The formula for it is:

$$P_{\text{out}}(x, y) = P_{\text{in}}(\text{round}(x), \text{round}(y)) \quad (1)$$

The bilinear interpolation method performs a linear weighted calculation on the pixel values of the adjacent 2x2 window in the source image. It can improve the visual effect and spatial continuity of the image, but may cause some image blurring and mixing of spectral features. The formula for it is:

$$P_{\text{out}}(x, y) = \frac{1}{(x_2 - x_1)(y_2 - y_1)} \left[P_{\text{in}}(x_1, y_1)(x_2 - x)(y_2 - y) + P_{\text{in}}(x_2, y_1)(x - x_1)(y_2 - y) + P_{\text{in}}(x_1, y_2)(x_2 - x)(y - y_1) + P_{\text{in}}(x_2, y_2)(x - x_1)(y - y_1) \right] \quad (2)$$

The cubic convolution method creates a new pixel value by taking a weighted average of the 16 nearest input pixels based on a 4x4 neighborhood window. It effectively balances the preservation of spatial details and edge smoothness. Additionally, it excels in both geometric accuracy and visual quality but processing time is increased due to complexity.

$$P_{\text{out}}(x, y) = \sum_{i=1}^2 \sum_{j=1}^2 P_{\text{in}}(x_i, y_j) k(x - x_i) k(y - y_j) \quad (3)$$

B. NDVI

$$\text{NDVI} = \frac{\text{NIR} - \text{RED}}{\text{NIR} + \text{RED}} \quad (4)$$

Fig. 3. NIR denotes the near-infrared band and RED denotes the red band of the sensor.

C. Models

Random Forest is an ensemble of decision trees that reduces overfitting by averaging many trees trained on bootstrap samples. It is well suited to tabular geospatial data with non-linear relationships. Support Vector Regression fits a flexible function that maximizes the margin around data points and can model smooth non-linear relations through kernels. Gradient boosting / XGBoost builds trees sequentially, each correcting errors of the previous ones, often achieving high accuracy on structured data.

VI. EXPERIMENT SETUP

A. Data Collection

UAV imagery was collected over a commercial strawberry field in Pomona, California, on 27 February 2025 using a multispectral UAV sensor. Landsat surface reflectance imagery covering the study area and closest in time to the UAV acquisition (24 February 2025) was downloaded from the USGS EarthExplorer portal. A digital elevation model (DEM) covering the Landsat footprint will be obtained from OpenTopography and later used to derive terrain predictors (slope, aspect, ruggedness).

B. UAV and DEM Processing

The raw UAV imagery (GeoTIFF) will be imported into Drone2Map to generate an orthomosaic and a UAV-derived surface model (DSM) at native UAV resolution. Where needed, a ground DEM will be derived or obtained, and terrain attributes (slope, aspect, ruggedness) will be computed from the DEM/DSM in Drone2Map or a GIS and exported as rasters co-registered with the UAV orthomosaic.

The external OpenTopography DEM will be resampled and aligned to the Landsat grid. From this DEM, slope, aspect, and ruggedness will be derived so that terrain predictors are available consistently across the full Landsat scene.

C. Clipping and Resampling

The Landsat surface reflectance product will be imported into ArcGIS Pro, reprojected if necessary to match the UAV coordinate system, and clipped to the UAV orthomosaic extent to define the overlap region for training.

To search for a meaningful common resolution, several candidate ground sampling distances (e.g., 0.25 m, 0.5 m, 1 m, 2 m) between the UAV and Landsat native resolutions will be tested. For each candidate resolution:

- UAV data will be aggregated (downsampled) to the candidate resolution.

- Landsat bands will be resampled to the candidate resolution using one of the resampling methods (nearest neighbor, bilinear, cubic convolution).

This will produce matched UAV–Landsat–DEM stacks at multiple resolutions for later modeling.

D. Prediction Area

Because Landsat covers a much larger region than the UAV footprint, a shapefile will be created to define a broader area of interest (approximately a 10-mile buffer around the strawberry field). This shapefile will be used to subset Landsat bands and the OpenTopography DEM when generating fused predictions at the selected optimal resolution.

E. NDVI Computation

NDVI rasters will be computed for both UAV and Landsat using the raster calculator in ArcGIS Pro, from the appropriate red and near-infrared bands. These NDVI layers will be co-registered with their corresponding spectral and terrain rasters at each candidate resolution.

F. Feature Compilation

For each candidate resolution, all processed layers within the UAV–Landsat overlap—Landsat spectral bands, Landsat NDVI, DEM-derived predictors (slope, aspect, ruggedness), and aggregated UAV NDVI—will be stacked. For each pixel:

- Predictors will consist of Landsat bands, Landsat and UAV NDVI, and DEM predictors.
- The target will be the corresponding NDVI (or suitability index) aggregated to that resolution.

This will yield one tabular training dataset per tested resolution.

G. Sample and Model Training

From each resolution-specific stack, samples will be extracted at the pixel (or plot) level to form training and validation datasets. Pixels affected by clouds, shadows, or artifacts will be removed.

For each candidate resolution, several regression models will be evaluated, including XGBoost, LightGBM, Random Forest, and Support Vector Regression. These models will serve as base learners in a stacking ensemble. Each base model will be trained to map the predictor set (Landsat bands, NDVI, DEM predictors) to the target UAV NDVI. A meta-learner (e.g., Random Forest or Ridge regression) will be trained on out-of-fold predictions of the base learners to form the final stacked model.

Model performance at each resolution will be assessed using RMSE, MAE, and R^2 on validation data. By comparing these metrics across resolutions, the study will identify the coarsest resolution that achieves near-maximum predictive accuracy, balancing spatial sharpness and physical realism.

H. Prediction and Output Generation

The best-performing model and its corresponding resolution will then be applied to the full Landsat scene within the area-of-interest shapefile. For each pixel in that area of interest, Landsat bands, Landsat and UAV NDVI, and DEM predictors will be extracted and passed through the trained stacking ensemble to generate predicted high-resolution NDVI values.

These predictions will be written to a fused NDVI raster at the selected resolution and compared visually and quantitatively to the original Landsat NDVI. The fused product will be summarized over management zones or fields to support precision agriculture interpretation.

VII. TIMELINE

Week	Goal
1	Collect the data for the satellite DEM and process DEM
2	Include the spectral bands in the new raster
3	Implement model with a linear regression for test
4	Implement code for testing the more advanced models
5	Test and validation
6	Implement ensemble learning
7	Test and validation
8	Determine the best model and create new NDVI images
9	Compile charts and predictions.
10	Research Poster
11	Research Poster
12	Research Poster
13	Work on research paper: Methodology
14	Work on research paper: Experiment Validation/Results
15	Work on research paper: Conclusion

So far, I have been able to collect the data for UAV and Landsat. And I tested this general experiment for linear regression, but only had NDVI as a feature, so now I need to include spectral bands and DEM as features.

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